

CMPG 311 Database Project

Group 5

Inventory Management and Sales Tracking System(IMSTS)

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Lecture :

Prof Francis Lugayizi

A database solution for efficient stock control, supplier management and sales tracking in retail businesses

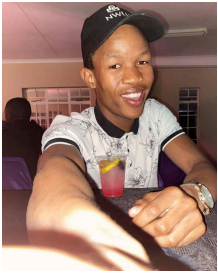
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Smart Inventory Management System : Phase 1

TEAM

Picture	Surname&Initials	Student_ID	Phone	GitHub_Link	Role	Contribution(%)
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	Mohapanelle KC	45737614	0784289290	https://github.com/Kamo471	Database Designer	100%
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1. Introduction

The inventory Management and Sales Tracking System(IMSTS) is a computerised solution developed to assist small retail businesses in managing their stock and sales activities efficiently. Many small business rely on manual methods such as notebooks or basic spreadsheets, which often results in errors, lost records and inefficiencies

This system is designed to improve how inventory is recorded, monitored and controlled. It will allow users to track products, manage supplier information, record sales and monitor stock levels in real time. By using a structured database, the system ensures that data is accurate, easily accessible and well organized

This document presents the team structure, system objectives, operational processes, existing challenges, and the specifications of the proposed database system

2. Team Roles and Responsibilities

To ensure the successful development of the Inventory Management and Sales Tracking System(IMSTS),each team member has been assigned a role based the responsibilities required for the project

These roles help ensure that the the database design,documentation,analysis and testing processes are completed effectively and in an organized manner

- **Project Manager - Riet OG**

Oversees the entire project and ensures that all phases are completed on time

Coordinates communication among team members

Monitor progress and ensures that project objective are achieved

Ensures that all deliverables meet the required standard

- **System Analyst - Semara K**

Examine the business problem and identifies system requirements

Helps align the project with the overall system objectives

Ensures that the proposed database solution addresses the needs of the business

- **Database Designer - Mohapanele K.C**

Ensures that the proposed database solution addresses the needs of the business

Identifies entities, attributes, keys, and relationships

Ensures that the database is logically structured and normalized

Ensures that the database is logically structured and normalized

- **Quality Assurance Analyst - Nemaguvhuni P**

Reviews the project content for errors, inconsistencies, and missing information

Checks whether the proposed design aligns with the project scope

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System Architect/Design Support - Mazibuko M.N

Assists in designing the overall system structure

Supports integration between different parts of the system

- **Documentation Specialist - Boyise N**

Compiles and formats the project document

Organizes all sections into a professional format

Prepares the final submission document

- **Technical Support - Mpofu T**

Prepares the final submission document

Assists with technical aspects of the project

Supports system design and implementation planning

3. Company Situation Analysis

Objectives(Mission)

The main goal of the business is to improve the management of inventory and sales by ensuring accurate, reliable, and accessible data.

The system aims to:

- maintain accurate stock records
- track product movement within the business
- improve sales recording and reporting
- manage supplier information effectively
- reduce stock shortages and overstocking
- support decision-making using reliable data
- monitor supplier activity and evaluate supplier performance based on delivery records

Company Operations

The business performs the following key operations, which must be supported by the database system:

→ Product Data Handling

The system manages product information within the business

- Store all product records
- Allow retrieval of product details
- Support updates to product information
- Allow safe deletion of products

→ Stock Control Operations

The system performs stock updates during operations

- Record stock increases (purchases)
- Record stock decreases (sales)
- Maintain current stock quantities at all times
- Prevent invalid stock values (e.g., negative quantities)

→ Supplier Data Management

The system manages supplier-related information

- Store supplier information
- Link suppliers to products and purchases
- Allow updates and retrieval of supplier information

→ Sales Processing

The current system records all sales transactions.

- Store each sale with product and quantity details
- Update inventory automatically
- Maintain historical sales records for reporting purposes

→ Purchase Processing

The system records incoming stock

- Store purchase transactions
- Link purchases to suppliers
- Update stock levels

→ User Management

System users already interact with the system

- Store user credentials securely
- Define roles and permissions are defined
- Track user activity

→ **Reporting Requirements**

The system supports reports and data retrieval

- Fast retrieval of inventory data
- Sales summaries can be generated
- Purchase summaries can be produced
- Low stock queries can be identified

→ **Audit and Tracking**

The system requires traceability of all activities

- Track all stock movements
- Record timestamps for transactions
- Track which user performed each actions

Detailed Company Organizational Structure

To ensure the effective operation of the Inventory Management and Sales Tracking System (IMSTS), a clear organizational structure has been defined. This structure outlines how data flows within the company, who is responsible for maintaining data accuracy, and the level of access each role has within the system

The organization is divided into three primary levels:

→ **Executive Management and Administration**

The General Manager is at the top of organizational structure and is responsible for strategic decision-making

- **Role in the System:** The Manager oversees the entire business lifecycle, they have full access to the database to pull high-level reports on total sales, profit margins, and inventory value.
- **Responsibility:** Use the system to look at long-term trends and authorize large orders from suppliers based on the data provided by the IMSTS.
- **Reporting Line:** All departments report directly to the General Manager.

→ Operational Level (Inventory Department)

This department manages stock and supplier-related activities.

The Inventory Manager:

- Responsible for managing product and supplier data
- Updates stock levels when new stock is received
- Verifies delivered stock against supplier records
- Has authority to add, update, and manage product and supplier information

Inventory Clerks:

- Assist with stock counting and verification
- Ensure physical stock matches system records
- Have read-only access to inventory data
- Support data accuracy and consistency

Data Control:

By centralizing the ability to "Update" and "Add" new products to a single department, the system eliminates the risk of duplicate entries or inconsistent naming conventions that plagued the previous manual system.

→ Front-Line Level (Sales Department)

This department handles customer transactions and sales recording.

Sales Supervisor:

- Monitors daily sales activities
- Approves returns and adjustments
- Oversees cashier performance

Sales Clerks:

- Record sales transactions
- Access product information such as prices and availability
- Have limited access restricted to sales operations

Access Control and Security:

- Sales Clerks cannot modify product prices or supplier information
- Sensitive data such as costs and supplier details is restricted

- This ensures data integrity and prevents unauthorized changes

Why This Structure Works

This structure ensures clear separation of responsibilities, reducing the risk of data inconsistency. By limiting who can modify critical data such as products and suppliers, the system maintains accuracy and reliability. This keeps the database clean, organized, and ready to scale as the business grows

Information Flow and Accountability

The system supports both top-down and bottom-up information flow:

- Sales Clerks generate transaction data
- Inventory staff manage stock and supplier data
- The General Manager reviews reports for decision-making

This ensures that every piece of data has a responsible owner, improving accountability and data integrity

To visualize how this structure supports the database, the following table outlines the functional permissions for each role within the IMSTS:

Role	Department	Key Database Responsibility	System Access Level
General Manager	Admin	Auditing & Strategic Reporting	Full Access (Superuser)
Inventory Manager	Operations	Purchasing & Stock Accuracy	Edit Access
Sales Supervisor	Sales	Transaction Oversight & Returns	Limited Edit
Sales Cashier	Sales	Day-to-Day Customer Sales	Create/Read (Sales Entry Only)

4. Problems and Constraints

Problems

There are problems one can encounter when using the manual system. These are a few of them:

- Recording inaccurate number of stock or sales.
- Records can easily be lost as we use manual systems like notebooks and basic spreadsheets. The data is not backed when making use of manual systems.
- The business may experience stock shortages due to the high demand of the product from consumers.
- The business may experience low sales due to overstocking or dissatisfaction of a certain product.
- Managing stock levels can really be challenging because staff members have to do stock taking manually to track whether sufficient stock is available.
- The business may also be unable to keep track of suppliers performance levels.

Constraints

- **Data Integrity Constraints**

This constraint assists by ensuring that data is consistent, reliable and accurate at all times

- **Stock Level Constraints**

This feature helps with minimizing stock level meaning that the business doesn't buy less or excess stock. To keep track of the stock levels the system will automatically update itself after every sale or restocking

- **Real-time update constraints**

The system should register the transactions immediately and effectively. This allows users to keep track of stock by being able to access the current stock level and sales information

- **Data Storage constraints**

The business must transition from making use of manual methods such as notebooks or basic spreadsheets to making use of a database system. This will avoid a loss of valuable information because we will create a database

system that will be able to backup data. The system will make use of a structured database to store data

- **User Authentication Constraints**

The database system will allow authorized personnel to access the system. Each and every employee of that company will have limited access to the system. For example, only staff members that are qualified can manage suppliers

- **Performance Constraints**

The system should be able to accept at least 100 concurrent users without crushing.

- **Supplier Management Constraints**

This constraint will assist the business to be able to keep track of the supplier's performance based on delivery records.

5.Database System Specifications

Objectives to Address Identified Problems

The proposed Inventory Management System database aims to improve inventory accuracy, accessibility, and operational efficiency by addressing the identified problems within the company.

- **Improve accuracy of stock and sales records**

The system will replace manual recording methods with a structured database to ensure that stock and sales information is captured accurately and consistently.

- **Prevent loss of important inventory records**

The database system will securely store inventory data electronically and support backup processes to reduce the risk of losing information stored in notebooks or spreadsheets.

- **Reduce stock shortages**

The system will monitor inventory levels and update stock automatically after each transaction so that low stock levels can be identified early.

- **Reduce overstocking and improve product management decisions**

The database will support tracking of stock movement and product performance to help the business maintain appropriate inventory levels.

- **Simplify stock level monitoring**

The system will automate stock tracking and reduce the need for manual stock-taking by providing updated inventory information in real time.

- **Improve supplier performance tracking**

The system will store supplier information and delivery records to help the business monitor supplier performance more effectively.

- **Ensure data consistency and reliability**

The database will enforce data integrity rules to maintain accurate and consistent inventory information at all times.

- **Provide real-time inventory updates**

The system will register transactions immediately after sales or restocking activities so that users can access current stock information when needed.

- **Secure inventory data through controlled access**

The system will implement user authentication and role-based access control to ensure that only authorised personnel can access or modify inventory information.

- **Support multiple users efficiently**

The system will be designed to allow many users to access inventory information simultaneously without affecting system performance

Information required for the database

The database must store and manage the following information:

- **Product Information**

The system must store details of all products, including:

- Product ID
- Product Name
- Description
- SKU or Barcode
- Category ID
- Price per Unit
- Cost Price
- Current Stock Quantity

→ **Supplier Information**

The system must store details of suppliers, including:

- Supplier ID
- Supplier Name
- Location
- Telephone Number
- Email Address

→ **Transactions information**

The system must store stock movement details, including:

- Transaction ID
- Product ID
- Transaction Type (Stock In / Stock Out / Adjustment)
- Quantity
- Date of Transaction
- Employee ID

→ **User Information**

The system must store user details, including:

- Employee ID
- Full Name
- Username
- Role

→ **Sales Information**

The system must store sales transaction details, including:

- Sale ID
- Sale Date
- Employee ID
- Total Amount

→ **Purchase Information**

The system must store purchase transaction details, including:

- Purchase Order ID
- Supplier ID
- Order Date

Clearly defined scope

The scope of the system defined what functions and activities are supported by the database

→ Supplier Management:

Contacts information for all suppliers should be managed and stored correctly including their Email Address, Cell Phone Numbers and their home address.

→ Stock Movement Tracking:

Each stock that comes in, and goes out, should be recorded as evidence to monitor the dates and the amounts (for monitoring the company progress in terms of profit and loss) This allows the business to monitor quantities and track inventory changes over time

→ Sales Processing:

The system must record all sales transactions, including the products sold, quantities, prices, and transaction dates. These records will ensure accurate tracking of stock leaving the business

→ Accountability:

The system should be able to Identify which Employee ID is responsible for particular stock transactions.

→ Product Cataloging:

All the list of products should be maintained including the price, category, name and SKU

Clearly defined Boundaries

The boundaries of the database system define what is included in the system, what is excluded, the operations supported, the users who may access the system, and how the data is protected

→ Scope boundaries

Scope boundaries define the type of data and activities that are included in and excluded from the database system

Included scope: the data stored and managed in the database will only be related to the retail business operational cycle

- **Product data** : Unique identifiers, descriptions, retail prices and categories
- **Stock levels** : Real time quantities, reorder and maximum stock levels.
- **Stock movement** : Records of stock entering and leaving the business, including stock received from suppliers, stock sold, returned stock, and stock adjustments
- **Sales data** : Information about products sold, quantities sold, selling prices, and dates of sale
- **Sales data** : Information about products sold, quantities sold, selling prices, and dates of sale
- **Purchase data** : Purchase records showing stock ordered from suppliers and stock received into the business

Excluded scope

The database will not include:

- Advanced accounting functions such as the general ledger processing
- payroll management
- tax filings
- fixed asset depreciation
- customer delivery tracking
- e-commerce order status management
- external banking or payment gateway integration

Functional boundary (operations supported by the database)

The database will support the following operations:

- storing and updating product information

- recording stock received from suppliers
- recording stock sold to customers
- maintaining current stock levels
- associating suppliers with purchase records
- recording daily business transactions
- supporting low-stock monitoring through stored data
- generating information for inventory and sales reporting

User boundary

User boundaries define who may access the database and the level of access they are allowed

- The admin and manager have access and permissions to the database
- Audit logs: tracking who modified or deleted certain records
- Inventory staff can update stock records, supplier information, and purchase-related data
- Sales Staff Can record and view sales transactions but cannot modify supplier or stock management records

Security boundary (how data is protected)

- Authentication: Unique user credentials with single logins
- Authorization and access control: Access to the database must be restricted according to user roles and responsibilities
- Manager has full access but cannot delete audit logs
- Sales clerk can only access the sales transactions but no access to supplier details and purchase orders